

Lecture 06

Programmatic Database Access with EDirect

What we previously learned

 An official website of the United States government [Here's how you know](#)

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

All Databases Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

[About the NCBI](#) | [Mission](#) | [Organization](#) | [NCBI News & Blog](#) | [Visit the Beta Homepage](#)

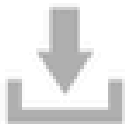
Submit

Deposit data or manuscripts into NCBI databases



Download

Transfer NCBI data to your computer



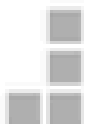
Learn

Find help documents, attend a class or watch a tutorial



Develop

Use NCBI APIs and code libraries to build applications



Analyze

Identify an NCBI tool for your data analysis task



Research

Explore NCBI research and collaborative projects



Popular Resources

- PubMed
- Bookshelf
- PubMed Central
- BLAST
- Nucleotide
- Genome
- SNP
- Gene
- Protein
- PubChem

NCBI News & Blog

BankIt Submitters: GenBank Submission Update

26 May 2026

As previously announced, major changes are being made to enhance user

Now Available: RefSeq Release 235

21 May 2026

RefSeq release 235 is now available online and from the FTP site! You can access RefSeq data

New Data Available! Access Hantavirus Sequences at NCBI

20 May 2026

What we previously learned

The screenshot shows the NCBI GenBank interface for the entry **Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence** (Accession: AB243863.1). A red box highlights the 'Send to' dropdown menu, which is open and shows the following options:

- ☒ Complete Record
- ☐ Gene Features
- Choose Destination**
 - ☒ File
 - ☐ Clipboard
 - ☐ Collections
 - ☐ Analysis Tool

Below these options, it says 'Download 1 item.' and 'Format' is set to 'FASTA'. There is a 'Show GI' checkbox and a 'Create File' button.

The main content area displays the following information:

LOCUS AB243863 1504 bp DNA linear BCT 08-DEC-2010
DEFINITION Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence.
ACCESSION AB243863
VERSION AB243863.1
KEYWORDS -
SOURCE Bacillus sp. YM9-109
ORGANISM [Bacillus sp. YM9-109](#)
Bacteria; Bacillati; Bacillota; Bacilli; Caryophanales; Bacillaceae; Bacillus.
REFERENCE 1
AUTHORS Goto,K., Asahara,M., Kasai,H. and Yokota,A.
TITLE Novel Bacillus species isolated from soil and sea water
JOURNAL Published Only in Database (2010)
2 (bases 1 to 1504)
AUTHORS Goto,K., Asahara,M., Kasai,H. and Yokota,A.
TITLE Direct Submission
JOURNAL Submitted (06-DEC-2005) Contact:Keiichi Goto Mitsui Norin Co.Ltd., Food Reseach Laboratories; 223-1, Miyabara, Fujiieda, Shizuoka, 426-0133 JAPAN, Fujiieda, Shizuoka 426-0144, Japan
FEATURES Location/Qualifiers
source
1..1504
/organism="Bacillus sp. YM9-109"
/mol_type="genomic DNA"
/strain="YM9-109"
/isolation_source="seawater in 40M depth"
/db_xref="taxon:361288"
/geo_loc_name="Japan:Kamaishi, Heita-Bay"
/lat_lon="39.15 N 141.54 E"
/collection_date="Nov-2003"
gene
1..1504
/gene="rrs"
1..1504
/gene="rrs"

Download FASTA

What if we want to download 100 files at a time??

The old method will consume lots of time

=> Use NCBI EDirect

NCBI Entrez Direct - NCBI EDirect

The Entrez Databases		
About 1,281,138,094 search results for "all[filter]"		
40 Databases, 1.3 billion records:		
Literature		
Books	337,145	books and reports
MeSH	253,765	ontology used for PubMed indexing
NLM Catalog	1,510,203	books, journals and more in the NLM Collections
PubMed	24,275,433	scientific & medical abstracts/citations
PubMed Central	3,242,724	full-text journal articles
Health		
ClinVar	125,230	human variations of clinical significance
dbGaP	166,367	genotype/phenotype interaction studies
GTR	35,697	genetic testing registry
MedGen	260,862	medical genetics literature and links
OMIM	23,757	online mendelian inheritance in man
PubMed Health	51,132	clinical effectiveness, disease and drug reports
Genomes		
Assembly	33,460	genomic assembly information
BioProject	137,524	biological projects providing data to NCBI
BioSample	2,813,447	descriptions of biological source materials
Clone	36,841,995	genomic and cDNA clones
dbVar	4,243,116	genome structural variation studies
Epigenomics	6,634	epigenomic studies and display tools
Genome	10,598	genome sequencing projects by organism
GSS	37,645,059	genome survey sequences
Nucleotide	153,484,042	DNA and RNA sequences
Probe	31,889,135	sequence-based probes and primers
SNP	394,165,245	short genetic variations
SRA	1,044,781	high-throughput DNA and RNA sequence read archive
Taxonomy	1,300,581	taxonomic classification and nomenclature catalog
Genes		
EST	75,694,136	expressed sequence tag sequences
Gene	18,027,510	collected information about gene loci
GEO DataSets	1,319,745	functional genomics studies
GEO Profiles	108,708,851	gene expression and molecular abundance profiles
HomoloGene	141,268	homologous gene sets for selected organisms
PopSet	210,603	sequence sets from phylogenetic and population studies
UniGene	6,473,284	clusters of expressed transcripts
Proteins		
Conserved Domains	49,955	conserved protein domains
Protein	152,126,214	protein sequences
Protein Clusters	820,546	sequence similarity-based protein clusters
Structure	103,194	experimentally-determined biomolecular structures
Chemicals		
BioSystems	647,610	molecular pathways with links to genes, proteins and chemicals
PubChem BioAssay	1,112,103	bioactivity screening studies
PubChem Compound	54,754,429	chemical information with structures, information and links
PubChem Substance	167,047,116	deposited substance and chemical information

- Entrez is a molecular biology database system that provides integrated access to nucleotide and protein sequence data, gene-centered and genomic mapping information, 3D structure data, PubMed MEDLINE, and more

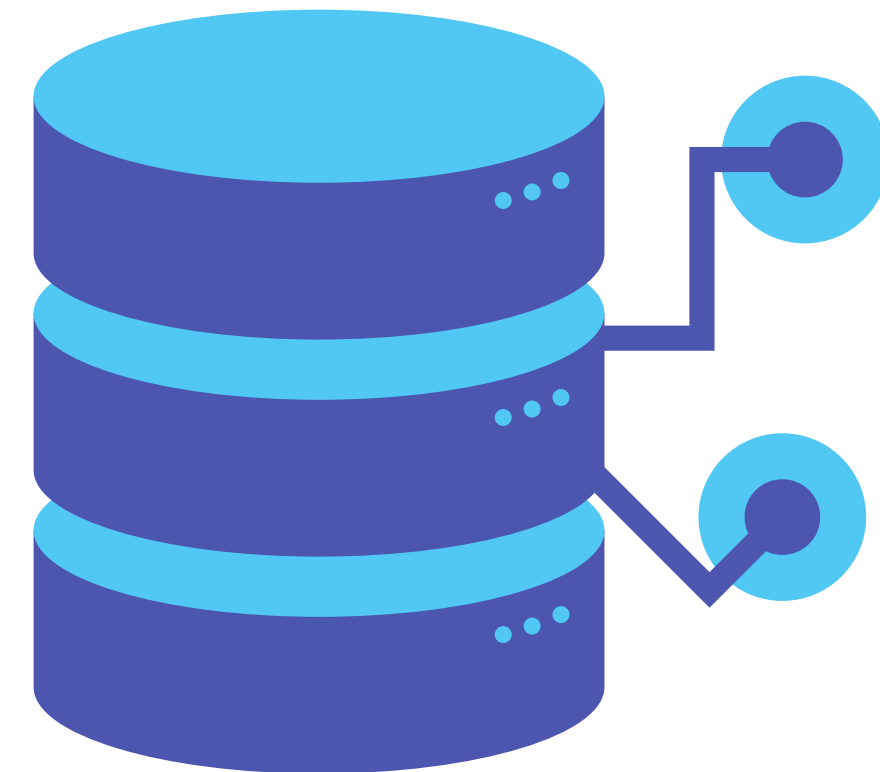
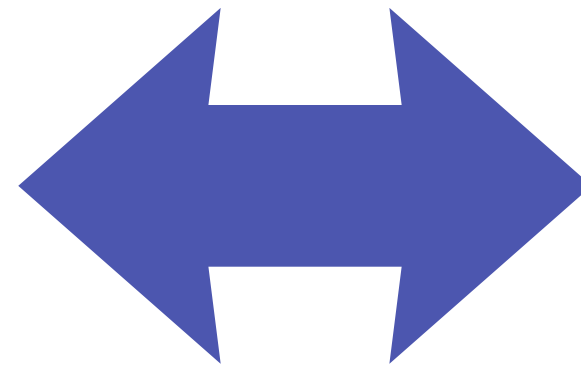
NCBI Entrez Direct - NCBI EDirect

NCBI Entrez Direct (EDirect) is a suite of command-line tools that allows you to interact with NCBI's massive biological databases directly from your terminal



Your terminal

NCBI EDirect



NCBI database

What is Entrez Database

Entrez Databases and UUIDs

Entrez Database	UID common name	E-utility Database Name			
BioProject	BioProject ID	bioproject			
BioSample	BioSample ID	biosample			
Biosystems	BSID	biosystems			
Books	Book ID	books	PopSet	PopSet ID	popset
Conserved Domains	PSSM-ID	cdd	Probe	Probe ID	probe
dbGaP	dbGaP ID	gap	Protein	GI number	protein
dbVar	dbVar ID	dbvar	Protein Clusters	Protein Cluster ID	proteinclusters
Epigenomics	Epigenomics ID	epigenomics	PubChem BioAssay	AID	pcassay
EST	GI number	nucast	PubChem Compound	CID	pccompound
Gene	Gene ID	gene	PubChem Substance	SID	pcsubstance
Genome	Genome ID	genome	PubMed	PMID	pubmed
GEO Datasets	GDS ID	gds	PubMed Central	PMCID	pmc
GEO Profiles	GEO ID	geoprofiles	SNP	rs number	snp
GSS	GI number	nucgss	SRA	SRA ID	sra
HomoloGene	HomoloGene ID	homologene	Structure	MMDB-ID	structure
MeSH	MeSH ID	mesh	Taxonomy	TaxID	taxonomy
NCBI C++ Toolkit	Toolkit ID	toolkit	UniGene	UniGene Cluster ID	unigene
NCBI Web Site	Web Site ID	ncbisearch			
NLM Catalog	NLM Catalog ID	nlmcatalog			
Nucleotide	GI number	nucore			

The E-utilities

<http://eutils.ncbi.nlm.nih.gov/entrez/eutils/eutil.fcgi?>

[esearch.fcgi?](#)

[egquery.fcgi?](#)

[esummary.fcgi?](#)

[einfo.fcgi?](#)

[elink.fcgi?](#)

[efetch.fcgi?](#)

[epost.fcgi?](#)

Requirement for using EDirect



Good internet



Web browser



Programming
language

NCBI Entrez Direct - NCBI EDirect

- EDirect is not one single command, but a suite of modular "navigation programs" that communicate with each other
- EDirect has many commands follow here for details:

<https://www.ncbi.nlm.nih.gov/books/NBK179288/>

EDirect Utilities

text query

Set of UIDs

Set of UIDs

Set of UIDs dbA

esearch

esummary

efetch

elink

einfo

xtract

Set of UIDs

Record summary

Data record

Set of UIDs dbB

EDirect Utilities - Web equivalent

Web action	E-utility equivalent
PubMed search	ESearch → ESummary
Click on title in search results	EFetch
Click to full text	ELink
Gene search	ESearch (→ ESummary/EFetch)
Find all RefSeq proteins for the gene	ELink

Basic utilities: esearch

BioSample

BioSample

rodent insulin

Your normal searching prompt

Search

Create alert Advanced

Organism
Customize ...

Attribute name
breed
cell line
cell type
collection date
cultivar
disease
geographic location
host
isolate
sex
strain
tissue
Customize ...

Sample type
Model organism or animal (213)
MIGS.eu (1)
MIMARKS.specimen (60)

Environmental
package
host-associated (60)

Access
Public (3,480)

Other
Used by SRA (2,781)

[Clear all](#)
[Show additional filters](#)

Summary ▾ 20 per page ▾ Sort by Has related data ▾ Send to: ▾ Filters: [Manage Filters](#)

Search results
Items: 1 to 20 of 3480

<< First < Prev Page 1 of 174 Next > Last >>

☐ [miR-10b_Knockdown_Day_8_1](#)

1. Identifiers: BioSample: SAMEA118336822; SRA: ERS24557390
Organism: Mus musculus strain: 129Sv
Accession: SAMEA118336822 ID: 57714626
[SRA](#)

☐ [Control_Day_8_1](#)

2. Identifiers: BioSample: SAMEA118336810; SRA: ERS24557378
Organism: Mus musculus strain: 129Sv
Accession: SAMEA118336810 ID: 57714538
[SRA](#)

☐ [Control_Day_5_1](#)

3. Identifiers: BioSample: SAMEA118336806; SRA: ERS24557374
Organism: Mus musculus strain: 129Sv
Accession: SAMEA118336806 ID: 57714426
[SRA](#)

☐ [Control_Day_8_2](#)

4. Identifiers: BioSample: SAMEA118336811; SRA: ERS24557379
Organism: Mus musculus strain: 129Sv
Accession: SAMEA118336811 ID: 57714045
[SRA](#)

☐ [Control_Day_8_3](#)

Find related data

Database: Select ▾

Find items

Search details

("Rodentia"[Organism] OR rodent[All Fields]) AND insulin[All Fields]

Search See more...

Recent activity

Turn Off Clear

rodent insulin (3480) BioSample See more...

The actual prompt for esearch

Filters: [Manage Filters](#)

Find related data

Database: Select ▾

Find items

Search details

("Rodentia"[Organism] OR rodent[All Fields]) AND insulin[All Fields]

Search

See more...

Recent activity

Turn Off Clear

rodent insulin (3480)

BioSample

See more...

Basic utilities: **esearch**

If you search NCBI for just the word "rodent", you will get millions of results. It will return the gene, proteins, papers written by people named rodent, and sequences from random bacteria where the author just mentioned the word in the notes.

To fix this, we use **Field Tags**. By putting a tag in brackets next to our word, we force NCBI to only look in that specific category.

Ex: Salmonella[Organism] AND mitochondrion [Title]
=> Search for article that has "mitochondrion" in it for Salmonella spp.

Filters: [Manage Filters](#)

Find related data

Database:

Find items

Search details

```
("Rodentia"[Organism] OR rodent[All Fields]) AND insulin[All Fields]
```

Search

[See more...](#)

Recent activity

[Turn Off](#) [Clear](#)

 rodent insulin (3480)

BioSample


```
<eSearchResult>
<Count>810</Count>
<RetMax>20</RetMax>
<RetStart>0</RetStart>
<IdList>
<Id>25303775</Id>
<Id>25295037</Id>
<Id>25276236</Id>
<Id>25250028</Id>
<Id>25221517</Id>
<Id>25165549</Id>
<Id>25143747</Id>
<Id>25114786</Id>
<Id>25101180</Id>
<Id>25071855</Id>
<Id>25045362</Id>
<Id>25003723</Id>
<Id>24963333</Id>
<Id>24920928</Id>
<Id>24904541</Id>
<Id>24868213</Id>
<Id>24811411</Id>
<Id>24791183</Id>
<Id>24779190</Id>
<Id>24648838</Id>
</IdList>
```

Total number of records found

Matching UIDs

Basic utilities: **esearch**

- esearch give a list of UIDs
- We can not know what are these samples, what to do with them

=> Use **esummary** to get there information

Basic utilities: **esummary**

```
(base) phuctran@schmoopy:~$ esearch -db nucleotide -query "insulin [GENE]"  
<ENTREZ_DIRECT>  
  <Db>nuccore</Db>  
  <WebEnv>MCID_6a1dec37c1e0503bfc0f6481</WebEnv>  
  <QueryKey>1</QueryKey>  
  <Count>6</Count>  
  <Step>1</Step>  
  <Elapsed>1</Elapsed>  
</ENTREZ_DIRECT>
```

No information or metadata

```
base) phuctran@schmoopy:~$ esearch -db nucleotide -query "insulin [GENE]" | esummary
?xml version="1.0" encoding="UTF-8" ?>
!DOCTYPE DocumentSummarySet>
DocumentSummarySet status="OK">
  <DocumentSummary>
    <Id>202471</Id>
    <Caption>M57671</Caption>
    <Title>Octodon degus insulin mRNA, complete cds</Title>
    <Extra>gi|202471|gb|M57671.1|OCOINS</Extra>
    <Gi>202471</Gi>
    <CreateDate>1991/03/01</CreateDate>
    <UpdateDate>1991/03/01</UpdateDate>
    <TaxId>10160</TaxId>
    <Slen>432</Slen>
    <Biomol>mRNA</Biomol>
    <MolType>rna</MolType>
    <Topology>linear</Topology>
    <SourceDb>insd</SourceDb>
    <SubType>tissue_type</SubType>
    <SubName>pancreas</SubName>
    <GeneticCode>1</GeneticCode>
    <Strand>ss</Strand>
    <Organism>Octodon degus</Organism>
    <Statistics>
      <Stat type="Length" count="432"/>
      <Stat type="Length" subtype="literal" count="432"/>
      <Stat type="all" count="5"/>
      <Stat type="blob_size" count="2763"/>
      <Stat type="cdregion" count="1"/>
      <Stat type="cdregion" subtype="CDS" count="1"/>
      <Stat type="gene" count="1"/>
      <Stat type="gene" subtype="Gene" count="1"/>
      <Stat type="imp" count="2"/>
      <Stat type="imp" subtype="polyA_signal" count="1"/>
      <Stat type="imp" subtype="polyA_site" count="1"/>
      <Stat type="org" count="1"/>
      <Stat type="pub" count="1"/>
      <Stat type="pub" subtype="PubMed" count="1"/>
      <Stat source="CDS" type="all" count="5"/>
      <Stat source="CDS" type="blast" count="5"/>
```

Basic utilities: **esummary**

Still unreadable!!!

Pipe to **xtract** to only get
the necessary field

Basic utilities: **esummary**

```
(base) phuctran@schmoopy:~$ esearch -db nucleotide -query "insulin [GENE]" | esummary | xtract -pattern DocumentSummary -element Caption,Slen,Title
M57671  432      Octodon degus insulin mRNA, complete cds
M26159  361      Rat insulin I 5' flanking region
NM_001204686  968      Aplysia californica insulin precursor (PIN), mRNA
NW_004798128  314614   Aplysia californica isolate F4 #8 unplaced genomic scaffold, AplCal3.0 scaffold00858, whole genome shotgun sequence
NTJ000000000  1213011  Eschrichtius robustus isolate AM-GW-M-EMB-4M-20130531, whole genome shotgun sequencing project
JWIN03000075  8536121  Camelus dromedarius breed African isolate Drom800 Contig74, whole genome shotgun sequence
```

Pipe to **xtract** to only get Caption, Slen, Title

Classwork 1: Use esearch and search for human COX1 gene from nucleotide database and summary it with only Caption, CreateDate, Title and view only the first 10 results

Basic utilities: **efetch**

- **esearch** only finds the ID numbers
- **efetch** retrieves the actual biological record
- **esearch** just gives us a count of how many things it found. To actually look at the DNA or read the paper, we need to hand those results over to **efetch** using the pipe symbol and use **efetch**



esearch will find the
title of the books



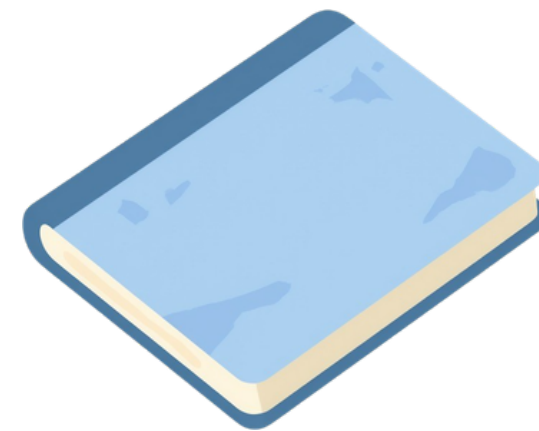
efetch will get those books
base on the title

Basic utilities: **efetch**

- EDirect is a combination of databases, so it has many type of file format for 1 UIDs. We need to use ***-format*** to get the desired format

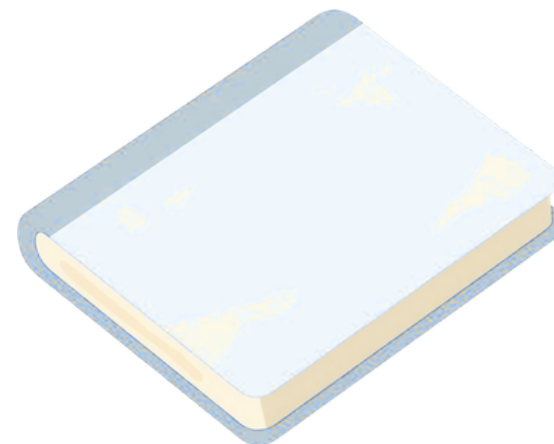


esearch will find the
title of the books



FASTA

efetch -format fasta



GB

efetch -format gb

Download multiple files with EDirect

- **esearch** find the IDs for your query
- **efetch** download all those IDs from **esearch** results without have to make a loop command

Classwork 2: Use **efetch** and download the first 20 human COX1 gene FASTA sequence from nucleotide database and export to a filename "human_cox1.fasta"

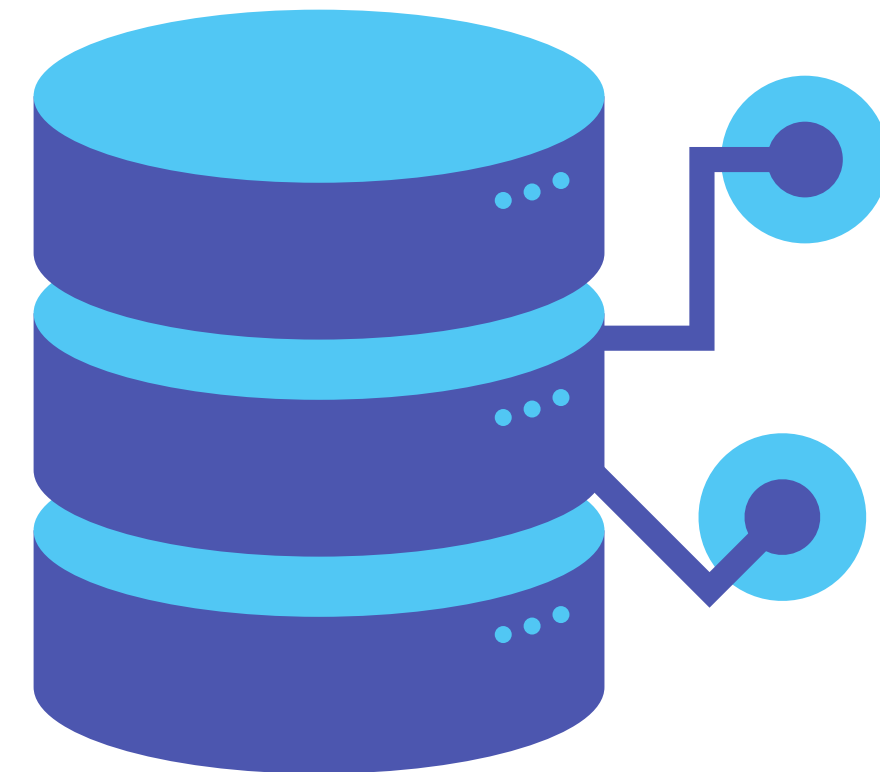
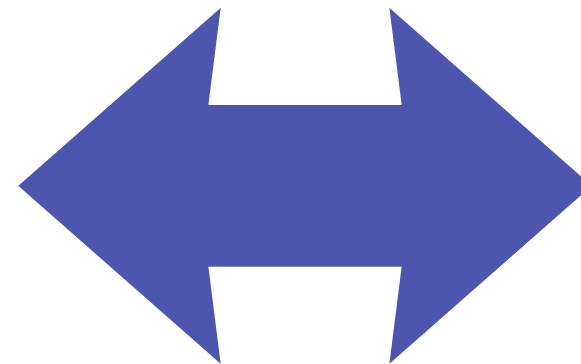
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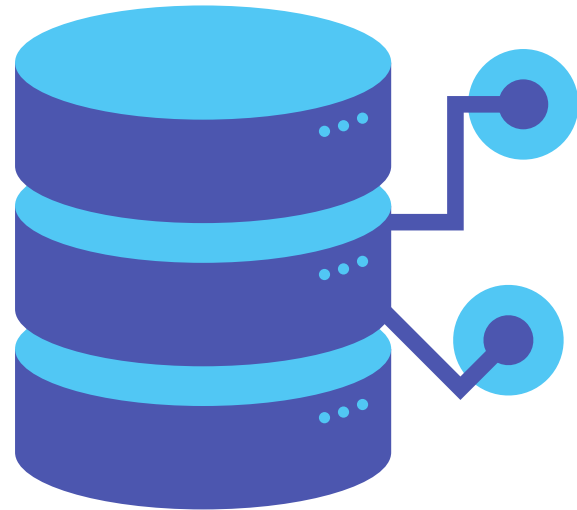
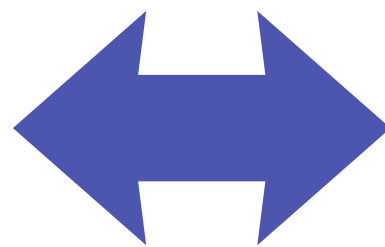
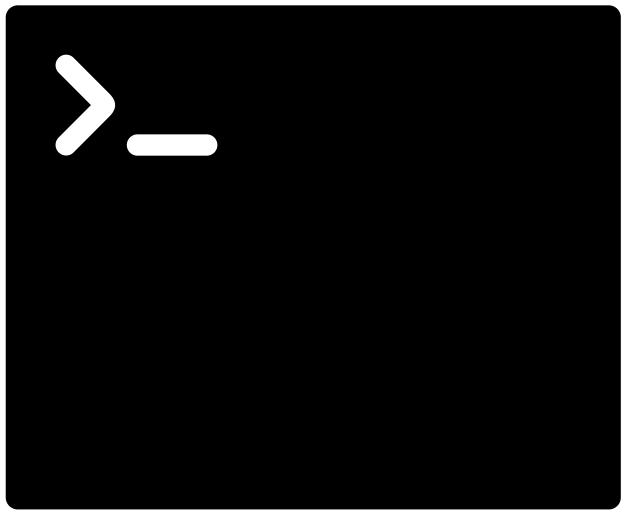
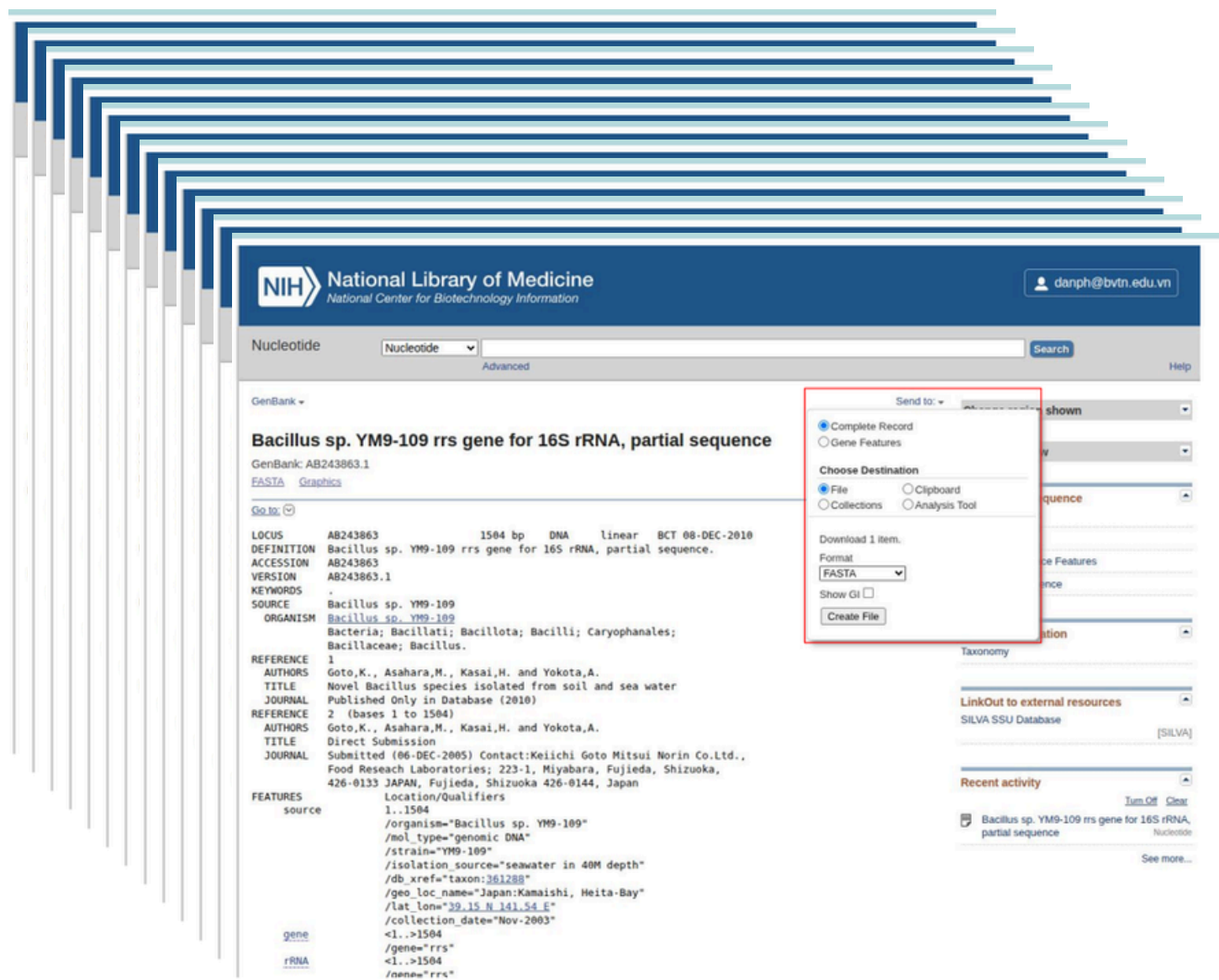
Your terminal

NCBI EDirect



NCBI database

Why using NCBI EDirect?



Clicking multiple times



Some command lines for multiple files